DATA STRUCTURES THROUGH C
(Common to EEE, CSE, IT, AIML & CSE-DS)

	L	T	P	C
II Semester				
Course Code:201ES2T07	3	0	0	3

Course Outcomes:

At the end of the Course, Student will be able to:

CO1: Interpret performance of a given algorithm
CO2: Apply searching and sorting techniques for problem solving
CO3: Make use of linear data structures for problem solving
CO4: Develop applications using Tree Data Structures.
CO5: Discuss basic concepts of Graphs to handle nonlinear data.
CO6: Solve problems using Graph Algorithms

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1	-	-	-	-	-	-	-	-	2
CO2	3	2	1	-	-	-	-	-	-	-	-	2
CO3	1	1	3	-	-	-	-	-	-	-	-	2
CO4	1	2	3	-	-	-	-	-	-	-	-	2
CO5	1	3	2	-	-	-	-	-	-	-	-	2
CO6	1	2	3	-	-	-	-	-	-	-	-	2

Unit - I
Data Structures –Definition, Classification and Operations on Data Structures, Pseudo code, Algorithm analysis, Time and Space Complexity. Searching: Linear search, Binary search. Sorting: Insertion Sort, Selection Sort, Exchange (Bubble Sort, Quick Sort),merging (Merge sort), distribution (Radix Sort) algorithms

Unit – II
Stacks: Introduction, Array Representation of Stacks, Operations and Implementation, Applications of Stacks-Reversing list, Infix to Postfix Conversion, Evaluating Postfix Expressions. Queues: Introduction, Array Representation of Queues, Operations and Implementation, Types of Queues: Circular Queues, Deques and Priority Queues, Application of Queues

Unit – III
Linked Lists: Introduction, Singly linked list, Operations on Singly Linked list - Insertion, Deletion and Searching, Doubly linked list - Insertion, Deletion, Circular linked list-Insertion, Deletion, Linked Representation of Stacks and Queues, Applications of Linked lists-Addition of Polynomials, Sparse Matrix Representation using Linked List

Unit – IV
Trees: Basic Terminology in Trees, Binary Trees-Properties, Representation of Binary Trees using Arrays and Linked lists, Traversing a Binary Tree(In-Order, Pre-Order,Post-Order). Binary Search Trees: Definition, Operations: Searching, Insertion, Deletion, Applications Expression Trees, Heap Sort, Balanced Binary Trees- AVL Trees, Insertion, Deletion and Rotations

Unit – V

Graphs: Introduction, Graph Terminology, Representation of Graphs-Adjacency Matrix and using Linked list, Graph Traversals(BFT & DFT),Applications-Minimum Spanning Tree Using Prims &Kruskals Algorithm, Dijkstra's Shortest Path, Warshall's Algorithm, Transitive Closure. (Algorithmic Concepts Only, No Programs required).

Text Books:

1. Data Structures Using C,Reema Thareja, Oxford University Press,2nd Edition.
2. Data Structures and Algorithm Analysis In C, Mark Allen Weiss,2nd Edition.

Reference Books:

1. Fundamentals of Data Structure in C, Horowitz,Sahni, Anderson Freed, University Press, 2nd Edition, 2008.
2. Data Structures, Richard F, Gilberg, Forouzan, Cengage Learning,2nd Edition.
3. Data Structures and Algorithms, G. A.V.Pai, TMH,2008.

Web Links:

1. http://nptel.ac.in/courses/106102064/
2. http://algs4.cs.princeton.edu/home/
3. https://faculty.washington.edu/jstraub/dsa/Master_2_7a.pdf
4. http://www.udacity.com/
5. http://www.courseera.com/